



GE VERNOVA

ADMS

16.25.1

Substation Builder User Guide

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Change History

Date	Change Description
May 30, 2023	ADMS 3.16: • Updated User Interface. • Updated Substation Builder Menus. • Updated Configuration. • Added Transformer Type Mapping. • Added Other Misc Settings. • Updated Saving a Configuration. • Updated Main Tab (Substation File Summary). • Updated User Preferences Tab. • Added Mapping Review Tab. • Updated Message Logs. • Editorial and formatting updates.
Oct 31, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Apr 1, 2024	ADMS 16.7.0: • Updated document format and styling.

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1. Introduction

The Substation Builder tool is a modeling tool that produces <Station>_subeditor.xml files that are used as an input to the Substation Editor tool, which produces Scada (formerly **e-terrascada**) displays and portions of the Scada model and the ADMS network topology model (DNOM model).

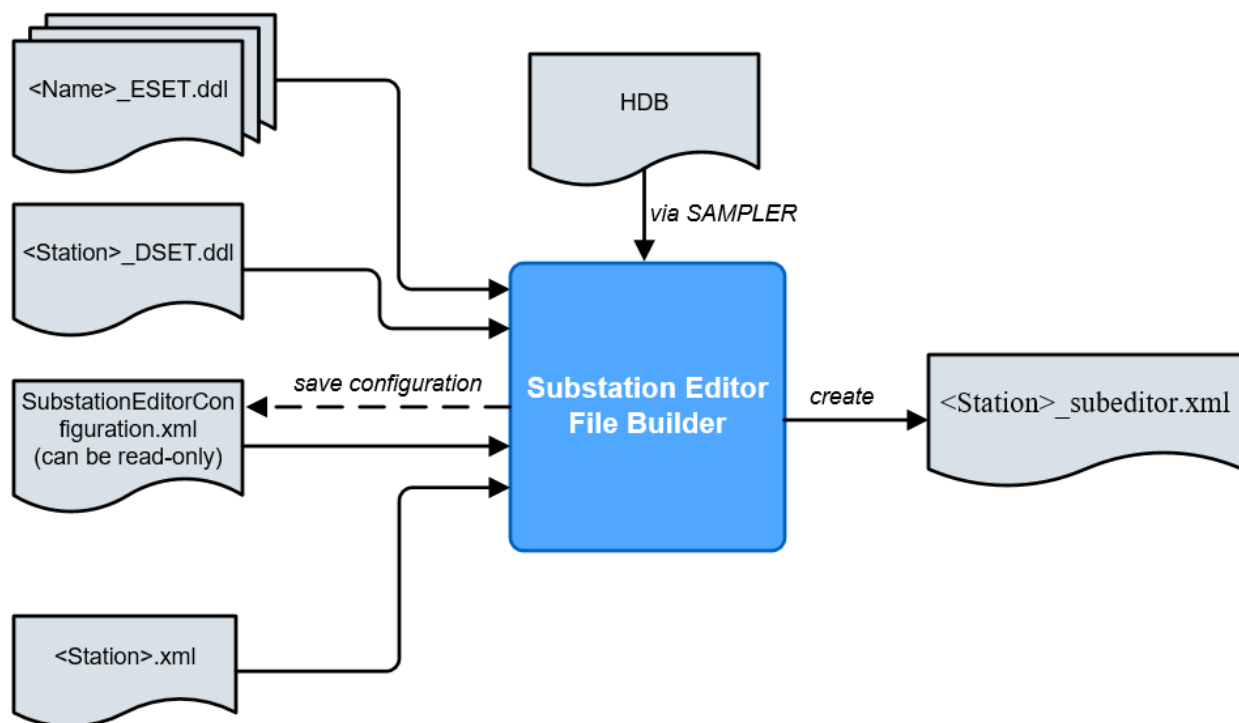


Figure 1. Substation Builder – Conceptual Overview

The Substation Builder tool uses the following files as inputs:

- **SubstationEditorConfiguration.XML:** This file preserves the modeling support constructs and configuration values that are common to all stations and views. For more details about configuring the Substation Editor tool, refer to the *Substation Editor User Guide*.

Any changes to the Substation Editor configuration are serialized to the SubstationEditorConfiguration.XML file.

- **UserPreferences.xml:** This file is serialized to and deserialized from the %UserProfile% directory. It preserves Substation Editor settings described in section "Set User Preferences" in the *Substation Editor User Guide*.
- **<Station>.xml or <Station>_SUB.csv:** Station externals files obtained from the asset management system (typically, GIS) and formatted as required by the Converter tool. For more information, refer to the *ADMS Modeling Reference Guide*.

These files are the primary source of information for the ADMS model, which are exported from the asset management system.

- **Scada Display Files (<Name>_ESET.DDL and <Station>_DSET.DDL):** These files are used for

extracting display information for generating <Station>_subeditor.xml files.

- **Habitat Database (HDB):** The Substation Builder tool connects to the Habitat database via Sampler to access the Scada connectivity information.
- **Substation Builder Configuration File:** This file contains configuration parameters that are used by the Substation Builder tool for generating the <station>_subeditor.xml files. For more details, refer to [Configuration](#).

The Substation Builder tool outputs the following files:

- **<Station>_subeditor.xml:** This file preserves the device, measurement, and graphic information used by the Substation Editor to produce the internals and placements files. In addition, this file preserves the modeling support constructs and configuration values that vary between stations or even between views.

2. User Interface

This chapter describes the controls and tabbed displays of the Substation Builder tool.

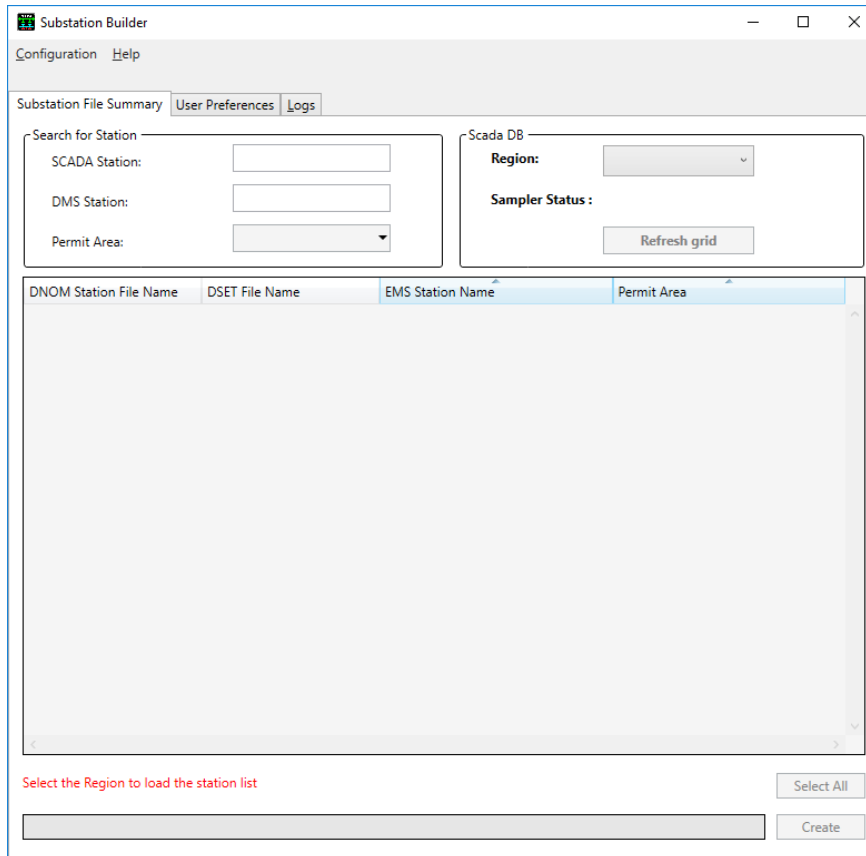


Figure 2. Substation Builder

2.1. Substation Builder Menus

The Substation Builder tool menu include the following options:

- **Configuration > Configuration Settings:** Opens the Configuration display. For more details, see [Configuration](#).
- **Configuration > Open Configuration file:** Reinitializes the application from a selected File Builder configuration.
- **Help > User Guide:** Opens the *Substation Builder User Guide*.
- **Help > About Substation Builder:** Shows the Substation Builder tool information.

2.1.1. Configuration

Clicking the Configuration menu option shows the Configuration window.

Configuration 1.0.0.00000

Point Type Mapping
Normally Opened

Region List
Regions

Feeder/Breaker Search
Prepend String
Append String

Transformer Type Mapping
Primary Winding
Secondary Winding
Transformer

Transformer List
☒ DMS_VOLTREG_034KV_LEFT
☒ DMS_VOLTREG_034KV_RIGHT
☒ DMS_VOLTREG_034KV_DOWN
☒ DMS_VOLTREG_034KV_UP
☒ DMS_VOLTREG_025KV_LEFT

Miscellaneous Settings
☐ Compare With Previously Created Internals
☒ Read Stations From Global File ☐ Update Permit Area Id From Global File

Parameter Mapping
Measurement Group
☒ Update Phase From Device
Update Phase for Analogs
Update Phase for Controls
Update Phase for statuses

Analog Map Name	Measurement Type	Unit	Phase	Base	Terminal	Picture Name	Voltage Measured Line to Line
3PF	PfMeasurementType	Base	ABC	0	0	LTC_RAISE	False
3PFTEST	PfMeasurementType	Base	ABC	0	0	LTC_RAISE	False
AAMP	MeasAmps	Base	A	1	0	ALG_LIM_NS_AMP_RECL	False
AMPA	MeasAmps	Base	A	1	0	ALG_LIM_NS_AMP_RECL	False
AMPB	MeasAmps	Base	B	1	0	ALG_LIM_NS_AMP_RECL	False
AMPC	MeasAmps	Base	C	1	0	ALG_LIM_NS_AMP_RECL	False
AMPG	MeasAmps	Base	C	1	0	ALG_LIM_NS_AMP_RECL	False

☒ Keep a backup before saving

Figure 3. Configuration Display

The Configuration display is used for changing the Substation Builder tool configuration properties.

Warning

The Substation Builder tool configuration parameters must not be modified when the tool is busy (processing a substation).

The Configuration display includes the following fields and controls:

- **Point Type Mapping:** Comma-separated list of normally open states in the SCADAMOM database.
- **Region List:** Comma-separated list for supported regions.
- **Feeder/Breaker Search:** This section enables mapping the feeder found in the station display file to that feeder in the substation external xml file based on the Prepend or Append String(s).
 - **Prepend/Append String:** Comma-separated list for configuring multiple Prepend/Append combinations. In the below example, a measurement on feeder "1234" (for example, STN.FDR.1234.ABC) will be associated with breaker names "1234**F**" or "**X**1234".

Feeder/Breaker Search

Prepend String

Append String

- **Transformer Type Mapping:** This section defines mapping for the primary winding symbol and secondary winding symbol to obtain the combined picture of primary and secondary windings. The configuration needs to be associated in the Substation Editor (SSE) for a new combined picture.
- **Miscellaneous Settings:**
 - **Compare With Previously Created Internals:** If selected, the tool compares the current DSET file with the previously created internal file to check whether there are any changes in the file. The tool only proceeds with creating Substation Editor XML files if the tool identifies any changes. Optionally, you can specify a comma separated "Exclusion List" elements (symbols and pictures), to exclude elements from the comparison, as shown in the [image](#).
 - **Read Stations from Global File:** If selected, the tool fetches station details from the global file instead of reading the data from the station external file and the SCADAMOM database to populate the Station Selection grid.

The Information in the global file is populated as follows:

SubName	0	INDEX	NAME_SUBNAME	EMSMAPNAME_SUBNAME	UNIQUEID_SUBNAME
SubName	1	1	RAVEN	RAVEN	8163417
SubName	1	2	SUN VALLEY	SUNVALLY	8224630
SubName	1	3	PINEHURST	PINEHRST	8227871

Where:

- SubName: The name of the group.
- Data Type: Type of data (0 header; 1 data).
- INDEX: The serial number for the data.
- NAME_SUBNAME: DNOM station name.
- EMSMAPNAME_SUBNAME: The EMS station name of the DNOM station.
- UNIQUEID_SUBNAME: The unique ID of the substation as defined in the Station External xml file.
- **Update Permit Area Id From Global File:** If checked, all substation elements' permit area IDs would be updated from the given original value (EMS_PERMIT) to the corresponding DMS_PERMIT value. The Information in the global file is populated as follows:

Name	Value	EMS_PERMIT	DMS_PERMIT
PERMIT_MAPPING	1	TOCBM	DOCBM

Name	Value	EMS_PERMIT	DMS_PERMIT
PERMIT_MAPPING	1	TOCEL	DOCEL
PERMIT_MAPPING	1	TOCLR	DOCLR

- **Use Scada Device Firmware Config:** A SCADARVersion custom field on the external SCADA devices indicates whether the device is DPF, FL, FISR, FISR LOV, LVM Ready, or not. The SCADAR versions are applied to the devices (Reclosers, Capacitors, Switchgear, and Regulators in the external) and indicate which measurements will be mapped to that device. The mapping implies which advanced applications the device can be used for.
 - **DPF Ready:** On an external SCADA device, if the SCADARVersion value implies that DPF is ready, then for MeasAmps, MeasVolts, MeasWatts, and MeasVARs analog measurements the DNAF flag is set to true or else it is set to false.
 - **FL Ready:** On an external SCADA device, if the SCADARVersion value implies that FL is ready, then for MeasAmps, MeasVolts, MeasWatts, and MeasVARs analog measurements the DNAF flag is set to true or else it is set to false.
 - **FISR LOV Ready:** On an external SCADA device, if the SCADARVersion value implies that FISR LOV is ready, then for MeasAmps, MeasVolts, MeasWatts, and MeasVARs analog measurements the DNAF flag is set to true or else it is set to false.
- **SCADA Device Firmware Config file:** This file contains a list of SCADARVersion values associated with external SCADA devices and its respective DPF Ready, FL Ready, FISR Ready, LVM Ready, and FISR LOV Ready values. The Subeditor file builder tool will read this csv file and will set the DNAF flag for analog measurements respectively. The file format is represented in the following table for better understanding:

SCADARVersion	0	DPF Ready	FL Ready	FISR	FISR LOV Ready	LVM Ready
RECL_651RA	1	T	T	F	F	T
RECL_652RA	1	F	F	T	T	T
RECL_653RA	1	T	F	F	T	F

- **Use MeasSide Custom Fields in Externals Config:** GISA will provide a Custom field named "Measurement_side" on the external SCADA devices. This field will contain the TO or FROM node of the device, depending on where the measurement node is placed. Based on the measurement node value, the device analog measurement's effective device value is set to the ID of the line connected to the device.
 - **Modeling measurement nodes for Reclosers, Switches, and Regulators:** If the "Measurement_side" value is set to "TO", then the device's respective analog measurements (MeasAmps, MeasWatts, and MeasVARs) effective device value is set to the line connected to the "TO" node of the device. If the "Measurement_side" value is set to "FROM", then the device's respective analog measurements (MeasAmps, MeasWatts, and MeasVARs) effective device value is set to the line connected to the "FROM" node of the device. If the device's analog measurement type is MeasVolts, then the device resultant node follows the below truth table:

MAPNAME	SFB Config Terminal	GISA Custom Field Node (Terminal Side)	Resultant Node
VAX	0	From	From
VAX	0	To	To
VAY	1	From	To
VAY	1	To	From

2.1.2. Parameter Mapping

This section shows measurement mapping for analogs, statuses, controls and setpoints.

You can add or delete entries using the context menu.

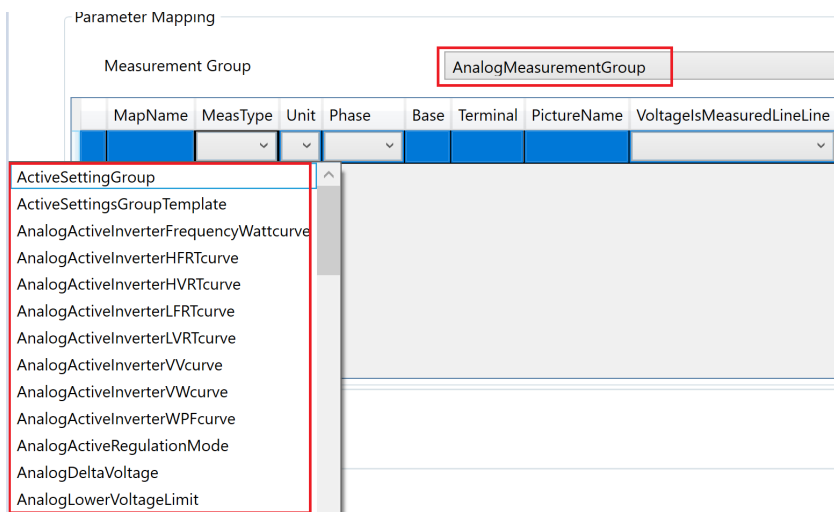
Note

The changes are not reflected on other application displays until the configuration is saved.

2.1.2.1. Mapping Options

Parameter mapping has the following options:

- **Measurement Group:** This list box contains the list of measurement groups. The Measurement Group grid shows the measurement mapping for the selected group.
- **Measurement Group and Meas Type Dropdown Filtering:** The Meas Type Dropdown in the Measurement Group Grid, displays only the measurement types specific to the selected measurement group.



- **Update Phase from Device:** When selected, the phase of the measurements (specified by comma separated values in the fields at the right) are read from the device instead of the mapping table. For example, The following image shows that for the analog map name **TAP** in

an external device (typically, the composite ID is %STATION%.%DEVTYPE%.%DEVICE%.TAP), the phase set uses the phase value of the external device instead of the "ABC" value in the mapping table.

When the Update Phase from Device option is checked, the fields for Analog, Status, and Control names are displayed and filled with comma-separated names of Analog, Status, and Control points for which the phase is read from the device.

Parameter Mapping

Measurement Group: AnalogMeasurementGroup

Update Phase for Analogs: TAP

☒ Update Phase From Device

Update Phase for Controls:

Update Phase for Statuses: STTS

Analog Map Name	Measurement Type	Unit	Phase	Base	Terminal	Picture Name	Voltage Measured Line to Line
SORT	AnalogActiveInverterLVRTcurve	Base	ABC	0	0	BRK_LOADSHED	False
TAP	XftapMeasurementType	Base	Unset	0	0	LTC_CONTROL_XFMR	False
UVLT	CapMeasurementType	Base	ABC	0	0	ALG_LIM_NS_AVLT	False
VAFA	MeasPhaseAngle	Base	A	0	0	ALG_LIM_MVAR	False
VAFB	MeasPhaseAngle	Base	B	0	0	ALG_LIM_MVAR	False
VAFC	MeasPhaseAngle	Base	C	0	0	ALG_LIM_MVAR	False

Figure 4. Parameter Mapping

2.1.2.2. Using Duplicate Map Name

If two rows have same values, the second row is not saved. However, the SCADA model may have the same map name for different measurements for different types of devices.

For example, the following image shows that "STTS" can be associated with the **switch status** measurement for a **switch** and with the **capacitor status** measurement for a **capacitor**. In this case, the "Device Group" field (comma separated values) distinguishes the two "STTS" rows. In the provided example, the backend SCADA model has all capacitors under "CAP or LCAP" device group or device type (for example, composite ID for the "STTS" measurement of the capacitor 1234 could be STATION.CAP.1234.STTS while composite ID for the switch 5678 could be STATION.SW.5678.STTS). Therefore, the first STTS has CapMeasurementType and the second has SwitchStatusMeasurementType.

In rare case, even with having the same device group, another device (for example, a fuse) might have a measurement "STTS". In this case, the DNOM class field could be specified to further specialize the measurement point.

Note

Duplicate map name is a relatively rare situation for most of SCADA models.

Parameter Mapping

Measurement Group: StatusMeasurementGroup

Point Name	Measurement Type	Phase	Picture Name	Device Group	Applies only for Dnom Class
STTS	CapMeasurementType	ABC	DSC_SWITCH	CAP,LCAP	
STTS	SwitchStatusMeasurementType	ABC	DSC_SWITCH		
SWMD	RecloserOpMode	ABC	DSC_SWITCH		
TGND	FaultStatus	G	FAULT_STATUS		
TRDP	DeviceSupervisoryCutout	ABC	DSC_SWITCH		

Figure 5. Valid Duplicate Map Name

2.1.2.3. Using Parent Fields

Controls and Setpoints are always associated with the Status and Analog measurement groups, respectively. If a Control Map Name is not unique, the Point (Parent) name could be specified to distinguish the map name from the other control with the same name.

The following image shows an example when a control "ON" is associated with a switching device, while another control "ON" is associated with a capacitor. The controls are distinguished using the "Point Name", which optionally specifies the point the controls belong to.

If Point (Parent) names are the same (rare case), you can use the "Applies to DNOM Class" field to distinguish the controls.

Parameter Mapping

Measurement Group: CtrlMeasurementGroup

☒ Update Phase From Device

Control Map Name	Point (Parent)	Measurement Type	Phase	Picture Name	Applies for Dnom Class
ON		CtrlCapStatusOnOutput	ABC	RO_RECL_CLOSE_DNOM_ONLY	
ON	STTS	CtrlSwitchStatusOpenOutput	ABC	RO_RECL_CLOSE_DNOM_ONLY	
REM		RemoteLocalDriver	ABC	RO_RECL_CLOSE_DNOM_ONLY	
RISE		CtrlRaiseTapOutput	ABC	RO_LTC_RAISE_3P_DNOM_ONLY	

Figure 6. Using Parent Field

i Note

Usually, SCADA measurement names are given unique names. For efficiency of the search logic, the device group, class, and parent fields can be used only when necessary.

2.1.3. Adding a New Measurement

To add a new measurement:

1. Right click a measurement row.

This shows a context menu with options to add or remove items.

2. Click Insert Row.

A new item is created just above the selected item. If the context menu is activated while no rows were selected, the new item is created at the end of the list.

3. Edit properties of the new item, according to the requirements.

If the application is closed using the Cancel or OK button or when the configuration is saved, all invalid items are removed.

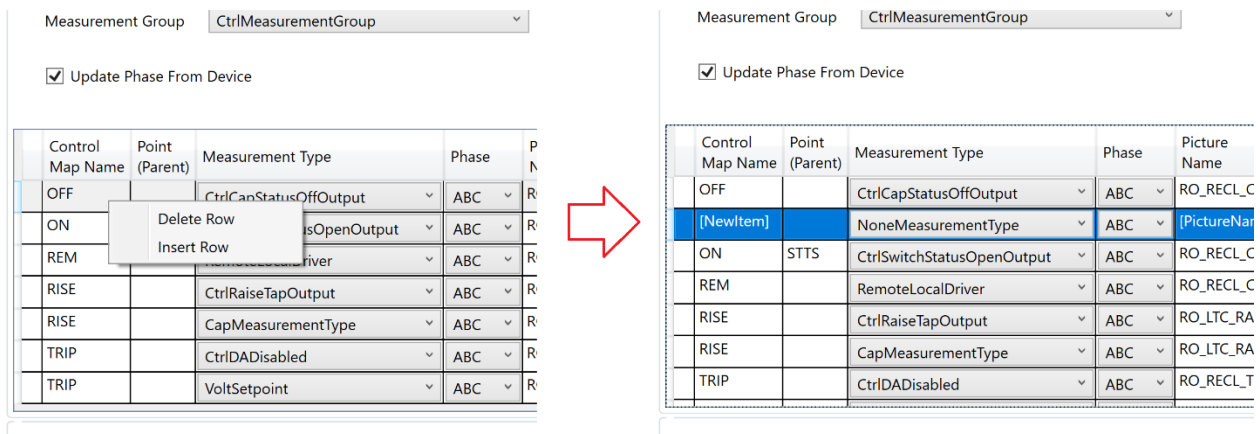
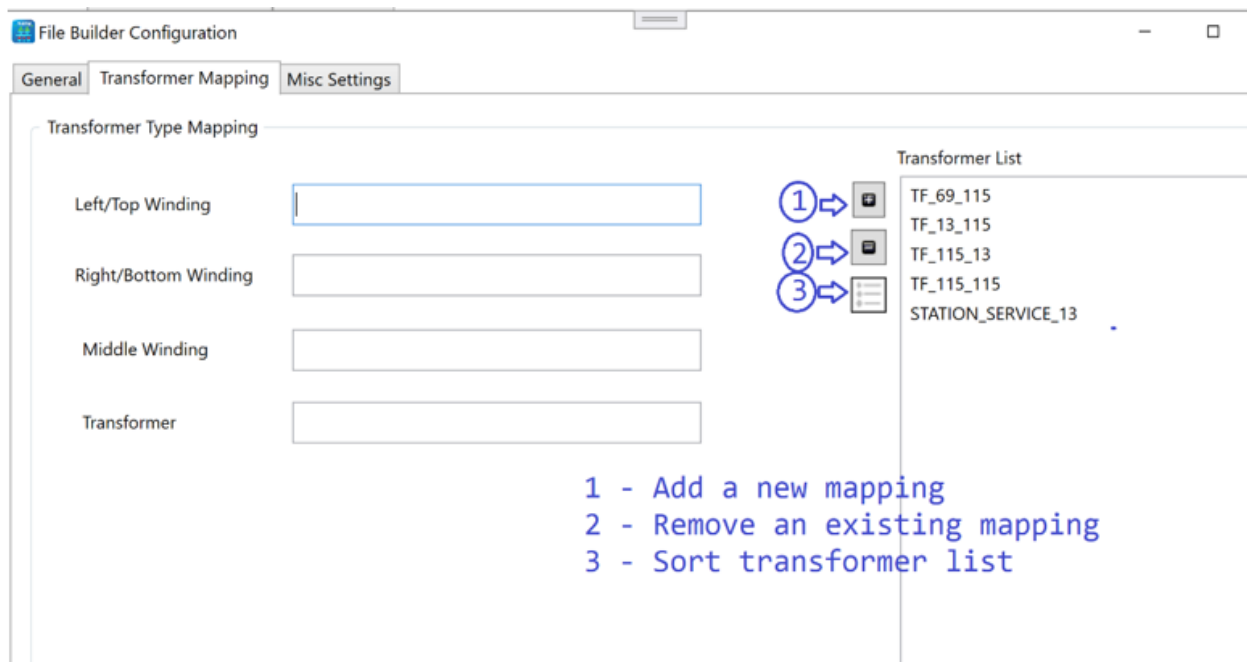


Figure 7. Adding a New Measurement

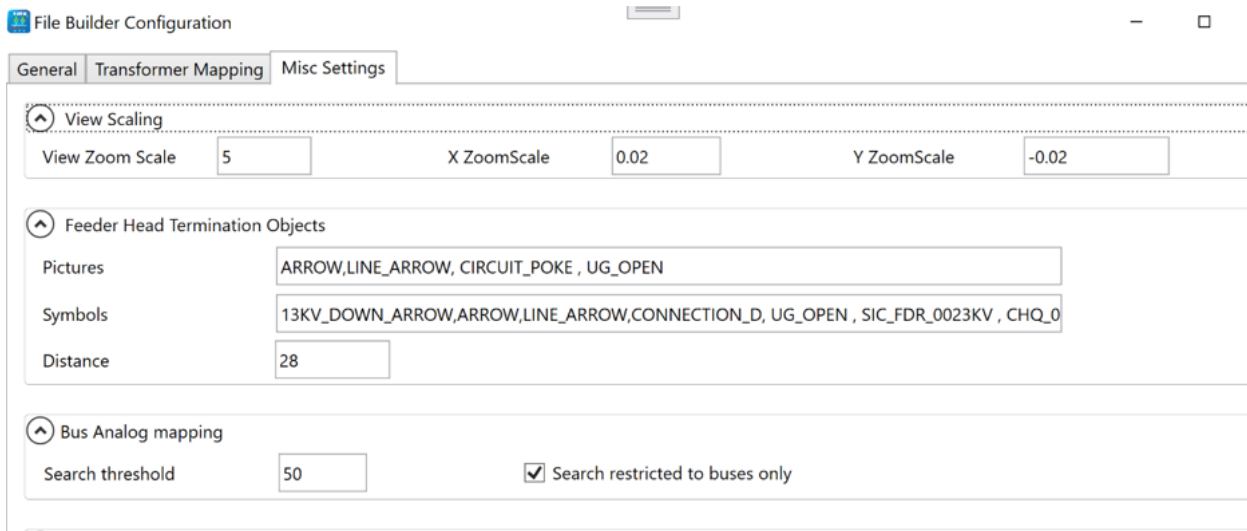
2.1.4. Transfer Type Mapping

This tab of the GUI defines mapping for the primary winding symbol and secondary winding symbol to obtain the combined picture of primary and secondary windings. The configuration needs to be associated in the Substation Editor (SSE) for a new combined picture.



2.1.5. Other Misc Settings

The following are the customer specific setting options in the Misc Settings tab:



- **View Scaling:** Sets zoom values of the final view.
- **Feeder Head Termination Objects:** Sets DSET symbol and picture names that are used to indicate termination of feeders (and start of a "ConnectionPoint"). The entries could be first few characters of the picture/symbol names.
 - **Distance:** The distance thresholds of the termination objects to the jumpers.
- **Bus Analog Mapping:** The distance thresholds used to search the bus objects to the analog

symbols in DSET. If, “Search Restrict to bus only” is unchecked, the proximity search would look for non-bus objects as well.

File Builder Configuration

General | **Transformer Mapping** | Misc Settings

DSET Processing

Line breaking tolerance: 3 ☒ Ignore rectangles ☒ Ignore polylines with empty gab names ☒ Clean unresolved connection points ☐ Remove floating jumpers

Exclude DSET Layer: EXTERNAL

Bus Gab Names: BUS,BUSS

Line Gab Names: DMS_LINE_034KV

Ignore Polyline Gab Name Pattern: BOUNDRY_LINE,TEXT_NOTES,NOTE_DISPLAY

Measurement Point Element: MEASPT_V Substation Label: DNOMSTATION_LABEL

Normal Interface Element Prefix: INTERFACE Study Interface element name: INTERFACE

Node Specifier (from-node)Element: NODE_SPECIFIER_D Node Specifier (to-node)Element: NODE_SPECIFIER_U

Interface Switch Element Name: DMS_SW_AT_INTERFACE

- **Line breaking tolerance:** The distance thresholds used in line breaking process.
- **Ignore Rectangles:** Ignore any rectangles in DSET.
- **Ignore poly lines with empty gab names:** If checked, empty gab named polylines in DSET are ignored.
- **Clean unresolved connection points:** Prior to final save, if there is a connection point with unresolved node id, the connection point is removed.
- **Remove floating jumpers:** If there is a jumper with empty FrNode or ToNode, the jumper is removed.
- **Exclude DEST Layer:** Comma-separated names of layers for exclusion in DSET processing.
- **Bus Gab Names / Line Gab Names:** Comma-separated gab names for Bus/Lines. Polyline with these Gab names are processed as Bus/Line.
- **Ignore Polyline Gab name Patterns:** Comma-separated names. If a polyline gab name contains any of the values, the polyline is ignored in DSET processing.
- **Measurement Point Element:** SSE Configured picture name for measurement point.
- **Station Label:** DSET picture for station label.
- **Interface Element Prefix:** The interface symbol name prefix (for example, “INTERFACE” – actual interface symbols looked for in code is “INTERFACE_H” for horizontal, “INTRFACE_V” for vertical oriented symbols).
- **Study Interface Element:** The study interface symbol name.
- **Node Specifiers (From/To):** The name of the Node Specifier Picture.

- **Interface SW Element name:** The name of the Switch element that is put in interface.

- **XF Scada ID:** The “DEVTYPE” parts of Transformer/Regulator measurements.
- **Substation Impedance File Processing:** Based on the Impedance file.
- **Use Substation Impedance CSV File:** Specifies whether to read the Substation Impedance or Reference Bus Interface Impedance from a Separate CSV File. The default value is False.
- **Negative Sequence Equals Positive Sequence:** Specifies whether to equate the Negative Sequence values to the Positive sequence values for Substation or Reference Bus Interface Impedance. The default value is False.
- **Impedance Data Provided for ReferenceBus:** Specifies whether the Impedance data provided in the Substation Impedance CSV File is for the Reference Bus, otherwise the given data will be considered for Substation Impedance. The default value is False.
 - **Use Node Search for RefBusInterface:** Specifies whether to locate the Reference Bus through the Node search for Impedance updation or Reference Bus is located using the Reference Bus name given in the Substation Impedance CSV File. The default Value is False.
 - **SCADA Device Firmware Config file:** This file contains a list of SCADARVersion values associated with external SCADA devices and its respective DPF Ready, FL Ready, FISIR Ready, LVM Ready, and FISIR LOV Ready values. The Subeditor file builder tool will read this csv file and will set the DNAF flag for analog measurements respectively. The file format is represented in the following table for better understanding:

SCADARVersion	0	DPF Ready	FL Ready	FISR	FISR LOV Ready	LVM Ready
RECL_651RA	1	T	T	F	F	T
RECL_652RA	1	F	F	T	T	T
RECL_653RA	1	T	F	F	T	F

- **Use Scada Device Firmware Config:** A SCADARVersion custom field on the external SCADA devices indicates whether the device is DPF, FL, FISR, FISR LOV, LVM Ready, or not. The SCADAR versions are applied to the devices (Reclosers, Capacitors, Switchgear, and Regulators in the external) and indicate which measurements will be mapped to that device. The mapping implies which advanced applications the device can be used for.
 - **DPF Ready:** On an external SCADA device, if the SCADARVersion value implies that DPF is ready, then for MeasAmps, MeasVolts, MeasWatts, and MeasVARs analog measurements the DNAF flag is set to true or else it is set to false.
 - **FL Ready:** On an external SCADA device, if the SCADARVersion value implies that FL is ready, then for MeasAmps, MeasVolts, MeasWatts, and MeasVARs analog measurements the DNAF flag is set to true or else it is set to false.
 - **FISR LOV Ready:** On an external SCADA device, if the SCADARVersion value implies that FISR LOV is ready, then for MeasAmps, MeasVolts, MeasWatts, and MeasVARs analog measurements the DNAF flag is set to true or else it is set to false.
- **Use MeasSide Custom Fields in Externals Config:** GISA will provide a Custom field named "Measurement_side" on the external SCADA devices. This field will contain the TO or FROM node of the device, depending on where the measurement node is placed. Based on the measurement node value, the device analog measurement's effective device value is set to the ID of the line connected to the device.
 - **Modeling measurement nodes for Reclosers, Switches, and Regulators:** If the "Measurement_side" value is set to "TO", then the device's respective analog measurements (MeasAmps, MeasWatts, and MeasVARs) effective device value is set to the line connected to the "TO" node of the device. If the "Measurement_side" value is set to "FROM", then the device's respective analog measurements (MeasAmps, MeasWatts, and MeasVARs) effective device value is set to the line connected to the "FROM" node of the device. If the device's analog measurement type is MeasVolts, then the device resultant node follows the below truth table:

2.1.6. Saving a Configuration

Until the configuration is saved, the changes are not visible on the other application displays.

To save configuration changes, click OK or Apply.

When the "Keep a backup before saving" option is selected, the last 5 changes (configurable) are saved to different files with the date and time appended to the existing name in the same folder as the source configuration.

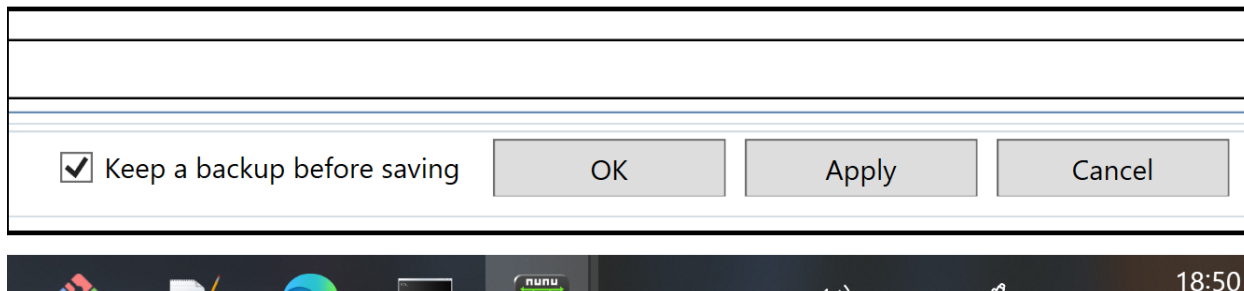


Figure 8. Saving a Configuration

2.2. Main Tab (Substation File Summary)

The Substation File Summary tab is used to perform most of the Substation Editor File Builder tool operations.

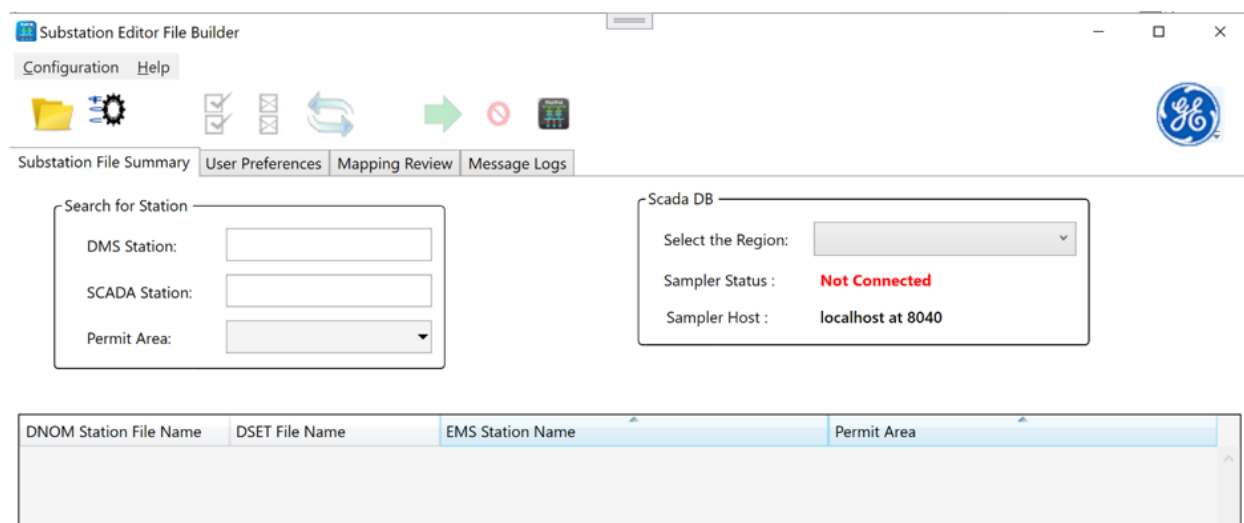


Figure 9. Substation Editor File Builder - Main Tab

The Substation File Summary tab includes the following fields and controls:

- **Search for Station:** This section is used to filter the substation list in the Station Selection grid.
 - **Search for Scada Station:** Filters the Station Selection grid based on the EMS station name.
 - **Search for DMS Station:** Filters the Station Selection grid based on the DMS station name.
 - **Search for Permit Area:** Filters the Station Selection grid based on the station Permit Area. This dropdown includes individual check boxes for the available Permit Area IDs. The station permit areas are obtained from the AREA_SUBSTN field in the SCADAMOM database.
- **Scada DB:** This section includes operations related to the Scada database.
 - **Select the Region:** The Substation Builder tool shows stations that are associated with the selected region in the Station Selection grid. This field list is populated through the Region List parameter on the Configuration display.

- **Sampler Status:** The Sampler application connection status (Connected or Disconnected). The connection parameters for the sampler application are configured on the Linker Tab in the Substation Editor Configuration window (for more details, refer to the "Linker" section in the *Substation Editor Installation Guide*).
- **Refresh the Grid:** Re-reads the Scada database and populates the Station Selection grid based on the selected region. When refreshing the grid, the Permit Area filter is reset to "ALL".
- **Station Selection Grid:** This grid is populated with the list of stations that are associated with the selected region.

DNOM Station File Name	DSET File Name	EMS Station Name	Permit Area
PEBBLE BEACH			
3LIONS	3LIONS_EDITOR_DSET	3LIONS	WST
BAKER	BAKER_EDITOR_DSET	BAKER	CTL
BETHPAGE	BETHPAGE_EDITOR_DSET	BETHPAGE	
COHO	COHO_EDITOR_DSET	COHO	WST
MARS	MARS_EDITOR_DSET	MARS	STH
PINE VALLEY	PINE VALLEY_EDITOR_DSET	PINEVALY	STH
TEST1	TEST1_EDITOR_DSET	TEST1	TEST1
TEST2	TEST2_EDITOR_DSET	TEST2	TEST2
WHISTLER		WHISTLER	CTL

The Station Selection grid includes the following fields and controls:

- **DNOM Station File Name:** DNOM station name in the Station External XML file.
- **DSET File Name:** Name of the DSET File which contains the display for the EMS station corresponding to the field <EMSName> in the Station External XML file.
- **EMS Station Name:** Name of the EMS station as fetched from the Scada database corresponding to the field <EMSName> in the Station External XML file.
- **Permit Area:** Permit area for the EMS station as fetched from the Scada database. This value is derived from the AREA_SUBSTN field in the SCADAMOM DB.
- **Select All:** Click to select all stations in the grid. Only valid stations are selected.
- **Create:** Click to create Substation Editor XML files for the selected substations.
- **Cancel:** Click to abandon Substation file creation process.

Features of the Station Selection grid:

- Stations in the grid can be selected using the Select All button or by clicking individual rows.
- Selected rows are highlighted in blue.
- Only the rows that passed all the validation checks can be selected.

Below are several validation fail examples:

- The Station External XML file does not have any entry for field <EMSName>. In this case only the Station External XML file name will be displayed; the other fields will be blank. The background color for EMS Station Name field will be red and the row will be non-

selectable.

- The Scada database does not have any station corresponding to the <EMSName> in the Station External XML file. In this case the DSET file name (if present) and the <EMSName> as per the Station External XML file will be populated, but the background color for EMS Station Name field will be red and the row will non-selectable.
- None of the DSET files in the DSET Directory field specified on the User Preferences tab have the display corresponding to <EMSName> in the Station External XML file.

The Progress Status bar shows the status of the current Substation Builder task.

2.3. User Preferences Tab

The User Preference tab contains configurations for the input and output Substation Builder folders.

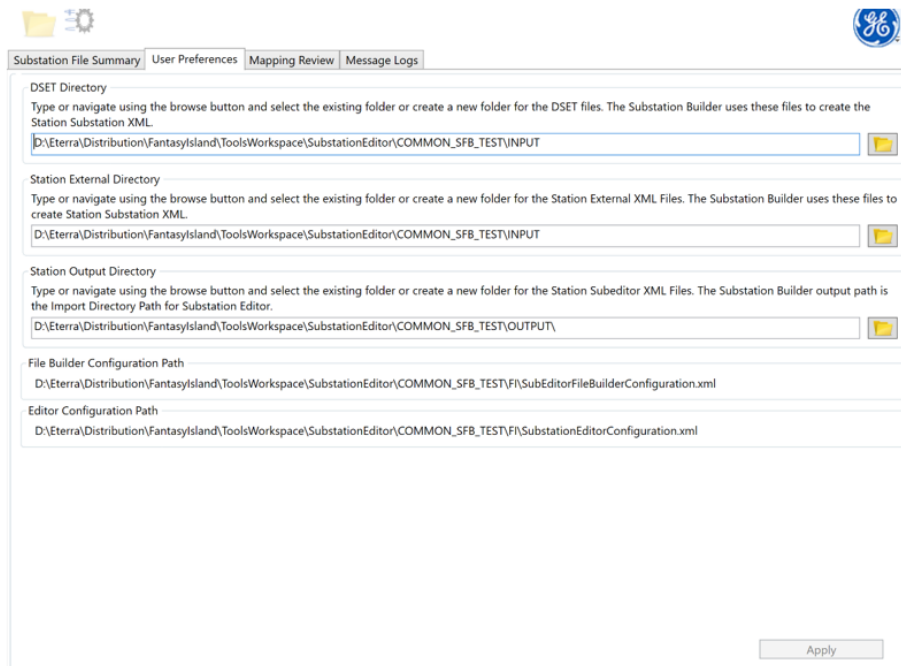


Figure 10. Substation Builder - User Preferences Tab

The User Preference tab contains the following options:

- **DSET Directory:** The path to the folder with station DSET files. The Substation Builder tool uses these files to create Substation Editor XML files. When changing the DSET directory path, the Substation Builder tool refreshes the Station Selection grid based on the new settings.
- **Station External Directory:** The path to the folder with Station External XML files. The Substation Builder tool uses these files to create the Substation Editor XML files. When changing the Station External directory path, the Substation Builder tool updates the Station Selection grid based on the new settings.

- **Station Output Directory:** The path to the folder for output Substation Editor XML files to be generated by the Substation Builder tool. This path is usually the "data" folder of the Substation Editor tool.
- **Apply Button:** Will save any change in DSET/ External/Output directories.
 - Besides above, it also displays current selected configurations.

2.4. Mapping Review Tab

The Mapping Review tab lists SCADA measurements with corresponding mapped DNOM object attributes. It also lists failed (null DNOM object) mappings. The station selection option is enabled only if there was a successful sub-station file creation.

File Builder Configuration

General Transformer Mapping Misc Settings

XF Scada ID

Scada Transformer Group ID VR,VREG,TF,XF,XFMR,LTC,XF_L,XF_H,TAP

Scada Regulator Group ID VR,VREG

Substation Impedance File Processing

SubstationImpedance CSV File

Type or navigate using the browse button and select the existing Substation Impedance CSV File.

☐ Use Substation ImpedanceCSV File ☐ Impedance Data Provided For ReferenceBus ☐ Negative Sequence Equals Positive Sequence ☐ Use Node Search For RefBusInterface

Scada Device Firmware Configuration

Scada Device Firmware Configuration CSV File

Type or navigate using the browse button and select the existing Scada Device FW Setting CSV file.

☐ Use Scada Device Firmware Configuration File ☐ Use Measurement Side Custom Fields in Externals

2.5. Message Logs

The Message Logs tab shows log entries for various operations performed by the tool.

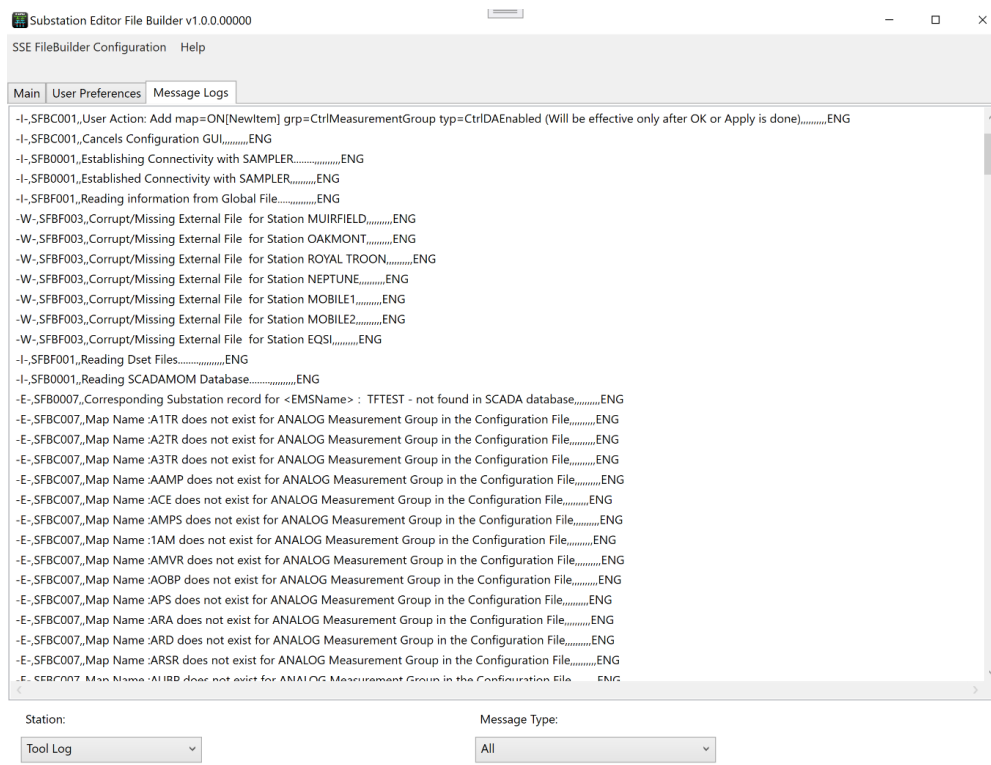


Figure 11. Substation Builder - Message Logs Tab

The following message types are supported:

- Information Messages (-I-)
- Warning Messages (-W-)
- Error Messages (-E-)

Use the following filters to filter the displayed messages:

- **Station:** Filters the log by the tool-related messages (using the Tool Log option) or station-related messages (using the <Station> option).
- **Message Type:** Filters the log by message type (Information, Warning, Errors, or All).

2.5.1. Extended Message Codes

The log messages typically look as follows:

```
"-T-....SFBX00N....This is a filebuilder log message"
```

Where, SFBX is one of the prefixes below. X=0 when the category is undetermined).

```
SFBF000 - File related issues (file read successfully or not found, corrupt etc)
SFBM000 - Mapping related issues (e.g. Analog is attached to a MW measurement)
SFB0000 - SFB Data related issues (e.g. Number of symbols read etc)
```

SFBT000 - SFB Tracing related issues

SFBC000 - SFB Configuration related issues (Config read/saved/edited etc)

00N is 001 (bit 0) for info, 00N is 003 (bit 0 and bit 1) for Warning, 00**N** is 007 (bit 0, bit 1 and bit 2) for Error.

i Note

Log error codes may have additional codes.

3. Using the Substation Builder

3.1. Starting the Tool

The Substation Builder tool is supplied as part of the ADMS Tools kit.

The Substation Builder tool requires the path to the Substation Editor configuration file as its input parameter. If the configuration file is not already provided in the executable shortcut's Target field, you may need to update the shortcut manually.

The default target value in the shortcut's properties dialog box is as follows:

```
Target D:\Eterra\Distribution\Tools\bin\SubEditorFileBuilder.exe /CONFIG
"%IDMS_ROOT%\ToolsWorkspace\SubstationEditor\SubstationEditorConfiguration.xml"
```

To start the tool, do one of the following:

- From the Windows Start menu, navigate to Eterra > Distribution > Tools > Modeling Tools > Substation File Builder.
- From within the Substation Editor application, click the SSE_FileBuilder_Callup button on the Substation Editor toolbar.

3.2. Command-Line Operations

The Substation Builder tool can also be run in command-line mode.

Table 1. Command-Line Arguments

Argument	Description
/CONFIG [PATH TO FILE]	Path to the Substation Editor configuration file.
/INCLUDE_LIST [PATH TO FILE]	Path to the CSV file that contains a list of substations to be processed by the Substation Builder tool to generate <Station>_subeditor.xml files for. The first row of the csv file is a header row (STATION_LIST). The subsequent rows contain station names. The example file contents are: <pre>STATION_LIST 3LIONS RAVEN VENUS</pre>
-OUTPUTDIR	Optional parameter. Path to a directory where the <Station>_subeditor.xml files are saved. By default, the file is saved in the Substation Builder tool directory.
-DSETDIR	Optional parameter. Path to a directory where station DSET files are located.

Argument	Description
-EXTDIR	Optional parameter. Path to a directory where station external XML files are located.

The Substation Builder tool command-line syntax is as follows:

```
SubEditorFileBuilder.exe /CONFIG [PATH TO FILE] /INCLUDE_LIST [PATH TO FILE]
```

For example:

```
SubEditorFileBuilder.exe /CONFIG
%IDMS_ROOT%\ToolsWorkspace\SubstationEditor\SubstationEditorConfiguration.xml /INCLUDE_LIST
%IDMS_ROOT%\ToolsWorkspace\SubstationEditor\SubstationList.csv
```

	A
1	STATION_LIST
2	3LIONS
3	TEST1
4	TEST2
5	
6	

Figure 12. Sample CSV File To Be Used For Batch Mode.